Prompt Self-Extension: A Training-Free Fix for Short-Utterance Failures in a CPU Persian Text-to-Speech Model

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Abstract*—Small continuous-latent text-to-speech (TTS) models now run in real time on a CPU, but short inputs are their weak spot. We study a 110M-parameter Persian model and make three points. (i) *Measurement:* the usual judge, Whisper, hallucinates on short audio; on real two-second recordings it fails 17% of correct clips. Scoring with two recognisers and counting a failure only when both agree, the true short-unit failure rate is 4–10%, five to ten times lower than Whisper alone reports. (ii) *Post-processing:* signal checks the engine gets for free (energy, duration, end-of-speech timing, latent self- and cross-similarity, syllable counts) predict which *units* are risky but cannot tell a failed take from a good take of the same unit, so retries without a recogniser recover little. (iii) *Pre-processing:* failures depend on how the voice prompt *ends*. We propose *prompt self-extension*: once per voice, the model reads one neutral sentence in that voice, and every later take is conditioned on the reference clip followed by that reading. Ending the prompt at the model's own completed sentence is the mechanism: cutting the extension mid-sentence multiplies failures by 3–10, and placing the clip last makes the model read the extension aloud. On held-out short sentences rendered through the full production pipeline with three voices, self-extension halves failed renders (24 → 12 of 900; consensus error 3.2% → 2.2%, 95% CI of the difference [−1.7, −0.4] points) and raises UTMOS for every voice; on three new, tightly cut real clips it cuts failures from 96 to 19 of 450, and a clip cut mid-clause goes from 44% to 3% error. The method needs no training and adds no per-request cost; its small speaker-similarity cost on some voices disappears when the extension take is chosen by speaker similarity. With it, the engine's retries can be dropped, halving the generation work while remaining more accurate.

***Index Terms*—text-to-speech, Persian, robustness, voice prompt, inference-time methods, low-resource**

I. Introduction

Continuous audio language models (CALM) generate speech as a sequence of continuous codec latents [1]; Pocket TTS distils such a model to a 100M-parameter student that runs faster than real time on a laptop CPU [2], and a community Persian student trained on 973 h is available [3]. We use it, exported to ONNX, as the speech engine of a Persian–English assistant. Its errors are not uniform: an assistant's short replies («باشه», «ممنون از پیامتان») are exactly where it fails, dropping final syllables or garbling the whole word.

Robustness of autoregressive TTS is usually bought with training: preference alignment and classifier-free guidance [4], fine-tuning on short phrases [5], or distilling ASR-verified best-of-N outputs back into the model [6]. These need the training pipeline, data with rights to use, and GPU time; for the released Persian student the 24-layer teacher it was distilled from is not even public. We ask what can be done at inference time only, and answer two questions:

1) **Can post-processing catch failures without a recogniser?** We build a labelled corpus of single takes and test every signal the engine sees for free. They identify risky units, not failed takes (Sec. IV).
2) **Can the input be changed so the takes do not fail?** We show that how the voice prompt ends drives short-unit failure and propose prompt self-extension, with an ordering rule that the model's training explains (Sec. V).

Answering both required fixing the measurement first: we show that a single ASR judge overstates short-unit failures several-fold and use a two-recogniser consensus instead (Sec. III).

II. Related work

Robust AR TTS. Hallucinated, repeated and dropped content is a known failure of autoregressive TTS [4], [7]. Asaria et al. remove most catastrophic failures with ASR self-verification and distillation [6]; GEPARD reports a systematic collapse on 1–2-word inputs and fixes it by fine-tuning [5]; LEMAS penalises repetition during decoding [8]; LoopGuard detects loops from token diversity in text LLMs [9]. All detectors we know of either use a recogniser or discrete tokens; a continuous-latent model has neither.

Context at inference time. For long-form speech, MagpieTTS-LF steers attention across chunks [10], Liu et al. post-train boundary-aware streaming [11], and Lim et al. find that the acoustic prompt must stay in context while speaker similarity still decays [12]. We study the prompt itself: its length and what it ends with.

Persian TTS. Open resources include ManaTTS [13], ParsVoice [14], homograph-aware G2P [15] and the PARSA-Bench audio benchmark [16].

III. System and failure analysis

Model. pocket-tts-farsi-v2 [3] is a 6-layer CALM student (110M parameters) over 24 kHz Mimi latents at 12.5 Hz. Text enters as Persian phonemes from a G2P model; English words are mapped to the same phoneme set. Each unit is generated from a fresh state: `[BOS, voice prompt, text, latents…]`. The voice prompt is a reference clip of at most

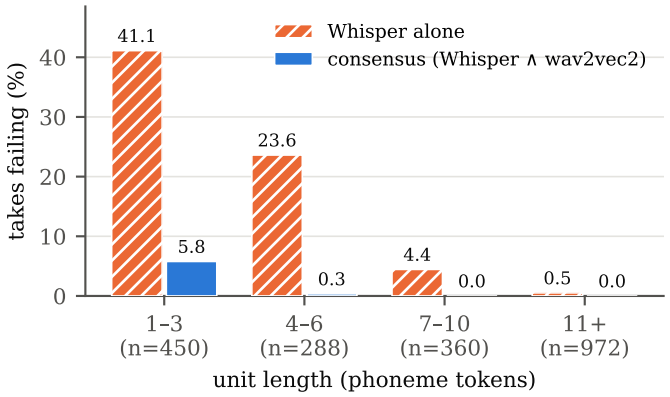


Fig. 1. Single-take failure rate (skeleton CER > 0.3) by unit length on the development corpus (2,070 Persian takes, three voices): Whisper alone vs. the two-recogniser consensus.

5 s; in training it was the audio *preceding* a random word boundary of the same utterance, with the words after the boundary as text.

Engine. Text is split into sentences and phrases, then into units of at most 30 phoneme tokens by dynamic programming over break costs (ezafe, verbs, «را»). Each unit is generated up to three times; a signal check (level, speech fraction, duration window, dead air) accepts a take or keeps the best of three.

Corpus. 110 development sentences (short replies, casual, formal, numeric, mixed, English, long) were planned exactly as in production, giving 127 units; each was generated six times with each of three voices (clips of 1.36 s, 2.0 s and 5.1 s), without retries: 2,286 takes. Each take is scored by consonant-skeleton CER: consonants mapped to sound classes, vowels dropped on both sides, because Persian script does not write short vowels.

Two judges. Whisper large-v3-turbo [17] (beam 5, digits suppressed) is a poor judge of short audio: on 76 *real* recordings of short Persian sentences (GPTInformal, CC0, mean 1.8 s) it charges skeleton CER 0.188 and fails 17% of them, often writing «موسیقی» ("music") for speech. A character-level wav2vec2-CTC recogniser for Persian (no language model) fails 10.5% of the same clips (0.115). On synthetic short takes the two judges correlate only 0.23. We therefore report both and use a *consensus* label: a take fails (CER > 0.3) only if both recognisers fail it, so a hallucination of either cannot fail a take. English units are judged by Whisper alone.

Findings. Under the consensus label 1.3% of the 2,070 Persian takes fail, all of them in units of at most six phoneme tokens and nearly all in units of 1–3 tokens (5.8% of those takes; none of the 972 takes of more than ten tokens). Whisper alone would report 12.3% overall and 41% for 1–3-token units (Fig. 1). Most failures are deletions (19 of 27), and the frames generated after end-of-speech are silent: the model, not the trimming, drops the ending. Short-unit error depends on the voice prompt (0.060 / 0.027 / 0.016 for the 1.36 / 2.0 / 5.1 s clips), and the two clips that ship with the engine end 0.02 s and 0.07 s after their last phoneme.

TABLE I
RETRY POLICIES REPLAYED ON STORED TAKES (DEV, PERSIAN, 345 UNIT×VOICE GROUPS, CONSENSUS CER / FAILURE RATE)

policy	takes/unit	CER / fail
first take	1	.0235 / 0.9%
engine signal QC (3 max)	1.49	.0215 / 0.6%
ASR-free detector (3 max)	1.31	.0251 / 1.2%
longest of 4	4	.0236 / 0.9%
latent medoid of 3	3	.0253 / 1.7%
oracle of 3 / of 6	3 / 6	.0133 / 0.3%, .0114 / 0.3%

IV. Post-processing without a recogniser

A retry loop is only as good as its accept/reject test. We computed, for every take, the signals an engine has for free. *Signal QC* (the engine's rule): level, fraction of frames with speech energy, speech duration against a token-rate window, longest dead-air run. *Timing:* end-of-speech step per vowel of the target, speech seconds per vowel, the number of syllable nuclei (peaks of the 0.3–2.5 kHz envelope) against the number of vowels in the phonemes. *Latent recurrence:* the strongest off-diagonal stripe of the cosine self-similarity of the take's latents over speech frames (0.64 s window, lag ≥ 0.48 s), which a take that says a stretch twice produces; and the strongest stripe of its cross-similarity with the prompt's latents, which a take that replays the prompt produces. A logistic regression on these signals is trained on the development corpus and evaluated by case-grouped cross-validation.

Units, not takes. With consensus labels the combined detector reaches a case-grouped cross-validated AUC of 0.88, but its strongest input is the unit length itself (AUC 0.94 alone). Holding the unit fixed — over all pairs of a failed and a good take of the same unit and voice — the best single signals reach 0.66 (speech fraction) and 0.67 (end-of-speech step per vowel), the engine's rule 0.58, latent recurrence and prompt similarity 0.48–0.57. The signals know which units are hard, not which take failed. They are not blind in general: on 300 takes made with the clip placed after the extension (Sec. V), 95% of which fail (the model reads the prompt's sentence), the prompt cross-similarity separates them from normal takes of the same units with AUC 0.98. Natural failures of this engine are different — short deletions — and those the signals do not see. The engine's rule is also badly calibrated: it rejects 29% of takes, 96% of them good ones.

Selection. Table I replays retry policies on the six stored takes per unit. The engine's rule spends 49% more generation for 2.35% → 2.15% mean CER; keeping the longest take (deletions dominate) or the latent DTW medoid does not help; an oracle choosing among the same takes halves the error. The headroom exists, but finding it needs a recogniser. The held-out bench corpus (79 cases, four takes per unit, 1,632 Persian takes) repeats the picture: 0.9% of takes fail, 13 of 15 by deletion in units of ≤6 tokens; the dev-trained detector falls to AUC 0.78 and its threshold does not transfer; retries move the error from 2.89% to 2.75% against an oracle's 1.23%. Since the failures sit in short units, we turn to the input.

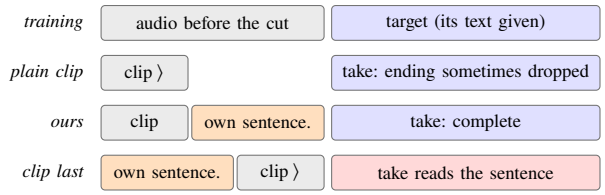


Fig. 2. Prompt layouts. In training the prompt is the audio preceding the target. A trimmed clip ends mid-utterance (⟩); prompt self-extension appends the model's own reading of a neutral sentence so the prompt ends at a completed sentence. With the clip last, the model replays the extension.

Algorithm 1 Prompt self-extension (once per voice)

Input: clip latents c ($|c| \le 5$ s), neutral sentence s, tries K

for $k = 1..K$ **do**
  $z_k \leftarrow$ generate(s | prompt c, seed k)
  $z_k \leftarrow$ speech span of z_k (+ one trailing frame)
end for
$e \leftarrow$ the z_k of median speech length {no recogniser}
$p \leftarrow [\, c$; last $(5\,\text{s} - |c|)$ of $e\,]$ {clip first, extension last}

Output: prompt p, used for every later take of this voice

V. Prompt self-extension

In training, the prompt is the audio that *precedes* the target in the same recording, and the target text starts after it. A reference clip that stops abruptly after its last phoneme — as cleaned clips do — asks the model to continue an unfinished utterance with words it was not given, and short targets, which leave the least room to recover, suffer most. We therefore extend the prompt with speech that the model itself produces in the target voice and that ends the way training prompts end: at a completed sentence (Fig. 2).

The extension is kept as the model's own (normalised) latents, converted back to the codec space the prompt is read in; decoding it to audio and re-encoding it worked less well in a Whisper-scored probe (0.286 vs 0.230 CER). Choosing the take with a recogniser instead (lowest CER of the K) was not consistently better (Table II); choosing it by speaker similarity is discussed below. The cost is K short generations when a voice is added, and none per request: the extended prompt is encoded once into the key–value cache that every unit already copies.

Order. With the clip placed *after* the extension the prompt again ends abruptly, and the model reads the extension sentence aloud instead of the text in 94–99% of takes under both judges: the take for the input «بله» begins with «امروز هوا کمی سرد», the extension's opening words. The order is therefore part of the method, not a detail.

Controls. Table II varies one thing at a time on the short development units (180 paired takes per cell, consensus label). More real speech with the same abrupt ending (the clip twice) does not help and hurts the male voice. The same extension cut after two thirds — same voice, same kind of latents, but ending mid-utterance — multiplies failures by 3–10. Teacher-forcing a carrier word before the text (continuation priming) has no effect, and conditioning each unit on the previous unit's last 2 s (to carry prosody across joins) adds nothing over the extension and, alone, lowers UTMOS by 0.11. Which sentence is read matters: of seven candidates, two (A, chosen first and fixed before any held-out test, and G) help both voices, whether the take is chosen with or without a recogniser; three hurt the male voice, among them a greeting (B) and two neutral statements (E, F). Neither the end-of-speech timing nor the engine's acceptance rate of the probe takes predicted which sentences work (rank correlations from −0.06 to −0.58, inconsistent across voices), so the sentence is a hyperparameter that needs a judge to choose; we report this as a limitation rather than tune it on test data. Sentence G, the other one that helped on development data, carries over only partly to the held-out set (male voice −1.5 points, CI [−3.3, +0.1]; female voice unchanged).

TABLE II
Prompt variants on short dev units: consensus CER (failed takes of 180)

prompt	male, 1.36 s clip	female, 2.0 s clip
clip (baseline)	.075 (18)	.033 (7)
clip twice	.136 (27)	.034 (8)
clip + ext. A	.045 (8)	.011 (2)
clip + ext. A, ASR-free take	.037 (4)	.021 (5)
clip + ext. A cut mid-sentence	.229 (48)	.333 (72)
ext. A + clip (clip last)	1.33 (169)	1.38 (179)
clip + carrier «خب» primed	.072 (16)	.027 (5)
clip + ext. B (greeting)	.148 (30)	.038 (7)
clip + ext. C (short replies)	.077 (16)	.032 (6)
clip + ext. D (formal)	.117 (28)	.036 (8)
clip + ext. A + B	.098 (17)	.026 (8)
clip + ext. E (neutral, past)	.102 (23)	.027 (6)
clip + ext. F (neutral, one clause)	.099 (23)	.034 (7)
clip + ext. G (neutral, habitual)	.043 (8)	.019 (4)

Held-out results. Two test sets were never used for any choice above. *Short:* 60 new sentences of 1–5 words (50 Persian, 5 English, 5 mixed), rendered through the full production path (front end, chunking, three-attempt signal retry, stitching). *Bench:* 79 mixed cases (prose, numbers, dates, English, code-mixed, acronyms, punctuation, 12 paragraphs). Table III pairs renders by case and seed. On short sentences, pooled over the three voices, failed renders halve (24 → 12 of 900 Persian renders; consensus error 3.2% → 2.2%, CI [−1.7, −0.4] points). The gain is significant for the 1.36 s clip that the engine ships as its default (6.0% → 3.9%) and small for the two voices that were already near the floor. UTMOS [18] rises for every voice (+0.09 to +0.17). Speaker similarity to the clip (WeSpeaker [19]) is unchanged for both male voices but drops by 0.03 for the female voice, whose reading of the extension drifts from the clip; this is the method's cost. On the bench, where most units are long, the final configuration also lowers the Persian consensus error (2.9% → 2.3%, CI [−1.0, −0.1] points), including numbers and dates; with Whisper alone over all 237 renders the change is not significant (5.65% → 5.05%), failed renders stay at 4, and English cases are unchanged because they keep the plain clip (applying the

Persian extension to English sentences had made acronym cases drift, 6.1% → 12.1%, n.s.). Speaker similarity drops slightly (−0.011). An unplanned effect is at chunk joins: the pitch reset between consecutive units of one sentence falls from 5.2 to 3.9 semitones and the utterance-final fall inside sentences from 1.9 to 1.3, i.e. units sound less like separate sentences.

New, tightly cut clips. The voices above were chosen before this work. As a final check we took three real 2.6 s recordings from GPTInformal (CC0) that the corpus cut 0.02 s after the last sound — one at the end of a sentence, one at a comma, one in the middle of a clause — and used them as voice prompts for the short set, building their extensions with Algorithm 1 and no further tuning. The mid-clause clip is nearly unusable on its own (consensus error 44%, 70 of 150 renders failed) and becomes ordinary with the extension (3.0%, 4 failed); the comma clip halves its error and the sentence-final clip improves slightly (Table III). Pooled, failures fall from 96 to 19 of 450. This is the prediction of the mechanism: the less finished the prompt's ending, the larger the gain.

Which take of the extension. The extension take matters as well as the sentence: two median-length choices of sentence A gave 4 and 13 failures on the male development units. Keeping instead the take whose WeSpeaker embedding is closest to the clip (still no recogniser) removes the speaker cost — pooled similarity +0.009 instead of −0.012, the female voice +0.010 instead of −0.030 — and gives the largest UTMOS gain (+0.18), but a smaller error reduction (−0.7 points, CI [−1.6, +0.1]). The choice of take is therefore a knob between intelligibility and voice fidelity; Table III reports both.

Retries become unnecessary. On short sentences the engine's three-take rule spends 2.1 generations per unit (its duration window rejects many good short takes), yet with the plain clip one take instead of three changes nothing measurable (6.0% → 5.9%). With the extension and a single take the error is 4.1% — −1.9 points against today's three-take default (CI [−3.3, −0.6]) — UTMOS is higher (+0.08), and the engine does 1.0 instead of 2.1 generations per unit (1.06 s instead of 1.88 s of CPU per short sentence in our runs). On a third test set, 40 assistant-style replies written for this check (1–15 words, five with English terms), the recommended configuration (extension for Persian sentences, one take) is not worse on any measure — consensus error 3.2% → 2.6%, Whisper failures 12 → 7 of 120, UTMOS and speaker similarity unchanged — while doing 28% fewer generations (1.38 → 1.00 per unit).

TABLE III
HELD-OUT RESULTS WITH THE FINAL CONFIGURATION (ASR-FREE EXTENSION), PAIRED RENDERS; CHANGE WITH 95% CASE-BOOTSTRAP CI

metric	clip	clip+ext	change [CI]
short set, 60 sentences × 6 seeds, male 1.36 s clip			
CER, consensus (fa, 300)	.060	.039	−.021 [−.036, −.008]
failed renders (fa, 300)	19	8	
CER, Whisper only (fa)	.316	.250	−.066 [−.107, −.024]
UTMOS (360)	3.01	3.12	+.12 [+.04, +.20]
speaker sim. (360)	.195	.193	−.001 [−.016, +.013]
same, female 2.0 s clip			
CER, consensus (fa, 300)	.020	.016	−.004 [−.009, +.002]
UTMOS (360)	3.42	3.59	+.17 [+.10, +.24]
speaker sim. (360)	.215	.185	−.030 [−.046, −.015]
same, male 5.1 s clip			
CER, consensus (fa, 300)	.017	.010	−.007 [−.017, +.004]
UTMOS (360)	3.29	3.38	+.09 [+.04, +.15]
speaker sim. (360)	.276	.270	−.006 [−.021, +.010]
all three voices pooled (fa, 900; UTMOS and speaker sim. over 1080)			
CER, consensus	.032	.022	−.011 [−.017, −.004]
failed renders	24	12	
UTMOS / speaker sim. change			+.13 / −.012
pooled, speaker-matched extension take			
CER, consensus	.032	.025	−.007 [−.016, +.001]
failed renders	24	20	
UTMOS / speaker sim. change			+.18 / +.009
short set, 3 seeds, three new real 2.6 s clips cut 0.02 s after speech			
CER, consensus, clip ends a sentence	.036	.020	−.017 [−.044, +.008]
CER, consensus, clip ends at a comma	.161	.088	−.073 [−.185, +.020]
CER, consensus, clip ends mid-clause	.441	.030	−.411 [−.546, −.288]
pooled failed renders (fa, 450)	96	19	−.167 [−.233, −.106]
UTMOS change (3 clips)			+.34, +.32, +.69
speaker sim. change (3 clips)			+.028, +.002, −.042
bench, 79 cases × 3 seeds, male 1.36 s clip, final configuration			
CER, consensus (fa, 165)	.029	.023	−.006 [−.010, −.001]
CER, Whisper (all 237)	.057	.051	−.006 [−.015, +.001]
failed renders (237)	4	4	
UTMOS (237)	3.28	3.28	−.01 [−.04, +.02]
speaker sim. (237)	.286	.276	−.011 [−.017, −.004]
pitch reset at joins (st)	5.25	3.92	−1.33 [−1.95, −0.77]
mid-sentence final fall (st)	1.93	1.28	−0.65 [−1.22, −0.05]

VI. Discussion and limitations

Why the ending matters. The training cut makes the prompt the audio *before* the target, so the model learns to continue from wherever the prompt stops. A prompt that stops on a completed sentence asks for a new one; a prompt that stops mid-phrase — a tightly trimmed clip, a clip after an extension, an extension cut short — asks the model to finish something it was not given the words for, and the result is a replay of the prompt or a short target with its ending dropped. Self-extension is one way to make every voice end the same, well-formed way without asking users for longer or pause-padded recordings; padding the clip with silence alone was inconsistent across voices in our probes.

Limitations. One model and six voice clips (three from one speaker); results may differ for discrete-token or attention-aligned TTS. Intelligibility is measured with recognisers, which is why we report two and their floor on real speech; a listening test is in progress [A/B pages prepared; results pending]. The extension sentence is a hyperparameter: of seven candidates two helped both voices, no simple property predicted which, and we fixed it on development data. English sentences, read through mapped Persian phonemes, did not benefit from a Persian extension (acronym cases drifted, 6.1% → 12.1%, n.s.); keeping the plain clip for sentences the front end marks as English removes this (6.1%, identical renders), so the engine extends the prompt for Persian and code-mixed sentences only. The extension changes the prosody of every unit, not only short ones; we measured joins and UTMOS but not long-form drift.

VII. Conclusion

Short inputs are where a small CPU TTS model fails, and a single ASR judge overstates how often. Signals available without a recogniser locate risky units but not failed takes, so retries without a recogniser mostly add cost. Making the voice prompt end the way training prompts end — with the model's own reading of one neutral sentence after the clip — halves failed short renders on held-out data, rescues clips that are cut mid-utterance, raises UTMOS, smooths chunk joins and costs nothing per request; it lets the engine drop its retries, halving the generation work while making fewer errors. Its price, a small loss of speaker similarity for one of three voices, disappears when the extension take is chosen by speaker similarity, at part of the intelligibility gain. Code, sentence sets and labels will be released with the paper [URL].

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